

Multivariable Proportional Hazards-Based Probabilistic Model for Bridge Deterioration Forecasting

Raka Goyal, A.M.ASCE¹; Matthew J. Whelan, A.M.ASCE²; and Tara L. Cavalline, M.ASCE³

Abstract: Duration-based deterioration modeling approaches have been investigated over the past two decades with a view toward utilizing their prognostic capabilities in accurately identifying maintenance needs and improving asset management under constrained budgets. However, despite significant advances in asset management systems during this time, implementation of such approaches for probabilistic infrastructure deterioration modeling has been limited, resulting in underutilization of their predictive potential. In this paper, a comprehensive framework based on a combination of the Cox proportional hazards method and Markovian theory is presented to develop multivariable deterioration models that probabilistically incorporate the effects of explanatory factors on deterioration over the complete life cycle of a bridge component. Both stationary (time-independent) and nonstationary (time-dependent) transition probability approaches are introduced and compared. Sample results from implementation of this framework on North Carolina's statewide bridge inspection database containing 35 years of data for more than 17,000 bridges are discussed. The predictive fidelity of the developed models is analyzed relative to the actually recorded condition ratings to demonstrate the effectiveness of these models in accurately forecasting deterioration.

DOI: 10.1061/(ASCE)IS.1943-555X.0000534. © 2020 American Society of Civil Engineers.

Author keywords: Bridge; Deterioration; Life cycle; Transition probability; Proportional hazards; Bridge management systems.

Introduction

Deterioration models are used to forecast the condition of key bridge components based on historical inspection records and are a critical component of bridge management systems (BMSs). Although deterioration modeling approaches have developed significantly since the inception of bridge management systems, increased accuracy of these models coupled with ease of implementation in the current BMS environments is essential for agencies to effectively forecast infrastructure needs and inform project prioritization under increasingly constrained budgets. Deterioration models employed in the earliest BMS frameworks were derived using purely deterministic methods based on averaging or linear regression (Hyman and Hughes 1983; Duncan and Johnston 2002). Subsequently, deterioration models based on Markovian state-based probabilistic approaches were developed (Jiang et al. 1988; Madanat et al. 1995; Frangopol et al. 2004), and these Markovian models were incorporated in the Pontis and Bridgit bridge management software (Golabi and Shepard 1997; Hawk and Small 1998). Although Pontis, now available commercially as the AASHTO-Ware Bridge Management (BrM 6.2), has been licensed by the majority of states in the United States, it is estimated that less than

20% of these states are using the deterioration modeling capabilities (Markow and Hyman 2009). This has been attributed to various reasons including lack of trained staff for using these models, lack of data analysis and preprocessing tools needed to generate the models, or lack of credibility of the available predictive models.

The traditional Markovian models, referred to previously, are based on the homogeneous or stationary Markov chain theory, in which the conditional probability of being in any future state is only dependent on the probability of the current state and is independent of the previous states. The predictive fidelity of Markovian models is dependent on the validity of the Markovian assumption and the accuracy of the matrix of transition probabilities associated with the rate of degradation across various condition states. Knowing the initial condition state of a bridge component, the probabilistic prediction of a future condition state is made using this transition probability matrix. The transition probabilities within the conventional Markovian model are stationary, or unchanging with respect to time, and are often developed based on expert opinion surveys of transportation engineers or calculated using simplified data-based methods such as the percentage prediction method or the expected value method, which uses linear regression of condition rating data (Butt et al. 1987; Jiang et al. 1988; Scherer and Glagola 1994; Wang et al. 1994; Madanat et al. 1995; Sobanjo and Thompson 2001; Frangopol et al. 2004; Sobanjo and Thompson 2011; Hatami and Morcous 2011; Hearn 2012). Although still in use, linear regression has since been considered inappropriate by researchers because the dependent variable, which is the condition rating in this case, is not continuous in nature as presumed under linear regression (Bulusu and Sinha 1997; Madanat and Ibrahim 1995; Madanat et al. 1995, 1997; Mishalani and Madanat 2002; Morcous et al. 2002). Moreover, the inherent assumption of time independence in these models is inconsistent with the time dependence of the deterioration process. The accuracy of the deterioration models also depends on correctly accounting for the effect of other explanatory factors besides age, such as material type, design type, geographical region, climate, and traffic characteristics. Historically, accounting

¹Research Associate, Dept. of Civil and Environmental Engineering, Univ. of North Carolina at Charlotte, Charlotte, NC 28223 (corresponding author). ORCID: <https://orcid.org/0000-0002-3645-4931>. Email: rgoyal2@uncc.edu

²Associate Professor, Dept. of Civil and Environmental Engineering, Univ. of North Carolina at Charlotte, Charlotte, NC 28223. ORCID: <https://orcid.org/0000-0001-6442-9496>. Email: mwhelan3@uncc.edu

³Associate Professor, Dept. of Engineering Technology and Construction Management, Univ. of North Carolina at Charlotte, Charlotte, NC 28223-0001. Email: tcavalline@uncc.edu

Note. This manuscript was submitted on October 9, 2017; approved on October 7, 2019; published online on March 3, 2020. Discussion period open until August 3, 2020; separate discussions must be submitted for individual papers. This paper is part of the *Journal of Infrastructure Systems*, © ASCE, ISSN 1076-0342.

for explanatory factors has been achieved by dividing the bridge population into categories based on these factors and developing deterioration models separately for each category. However, this categorization is frequently carried out on the basis of engineering judgment rather than statistical methods. Moreover, the extent of categorization is necessarily limited by the number of bridges required to produce statistically reliable deterioration models in each category at the lowest levels of the classification hierarchy.

Progressive improvements to the calculation of transition probabilities for state-based Markov chain models have been developed to address limitations of traditional approaches. The discrete nature of the dependent variable was first addressed through development of a Poisson regression model, which was later extended into a negative binomial regression model to relax the constraining assumption of equality of variance and mean in Poisson regression (Madanat and Ibrahim 1995). In addition to improving the predictive fidelity of the previous model, this approach also permitted the development of a relationship between deterioration and the various explanatory variables affecting it. By doing so, it eliminated the need for segmenting the bridge population into homogeneous groups so that the statistical advantage of having the entire data set for estimation was obtained. Ordered probit models developed around the same time also accounted for the ordinality of the dependent variable and the time dependence of the deterioration process (Madanat et al. 1995). Although the transition probabilities calculated using these models were found to be more accurate in prediction than prior models, these models were still considered deficient in their ability to address the two issues of heterogeneity and state dependence found in panel data, or longitudinal data (Bulusu and Sinha 1997; Madanat et al. 1997). Recent approaches to calculating the transition probabilities of stationary Markov chain models include a Metropolis-Hastings algorithm-based Markov chain Monte Carlo simulation technique used to calibrate state-based Markov deterioration models of Australian railway bridge components (Wellalage et al. 2015).

The issue of heterogeneity was addressed by developing the binary probit and random effects models as alternatives to the conventional Markov chain models. Although no appreciable difference was observed in the coefficient values of explanatory variables, these models were found to improve significantly the goodness of fit and predictive fidelity relative to the previous models (Madanat et al. 1997; Bulusu and Sinha 1997). Recently, the bridge deterioration models in the Indiana BMS were updated using the binary probit model (Moomen et al. 2016). This study also statistically evaluated the influence of explanatory variables on various bridge elements in developing the new models.

The issue of state dependence, however, remained unresolved. Madanat et al. (1997) found that state dependence was present and correlated heavily with the elapsed time in the condition state. However, once the effect of heterogeneity is accounted for, it is difficult to distinguish between the effects of time nonhomogeneity as captured in nonhomogeneous Markov chain models and true state dependence (Madanat et al. 1997). The growth of the historical condition rating databases over the past decade has supported the investigation of duration-based probabilistic approaches, which, unlike previous models, can account for the duration dependence of deterioration processes. Several studies illustrate the use of survival analysis in modeling transition time in an individual condition state before deterioration to a lower state (Ravirala and Grivas 1995; DeStefano and Grivas 1998; Ng and Moses 1998; Mauch and Madanat 2001; Mishalani and Madanat 2002; Agrawal et al. 2010). Some of these studies have demonstrated the use of multivariable survival functions to study the effect of explanatory factors on deterioration (Mauch and Madanat 2001; Mishalani and

Madanat 2002). Approaches based on parametric Weibull survival functions have been presented for calculating nonstationary transition probabilities of individual condition states (Ng and Moses 1998; Mishalani and Madanat 2002; Kallen and van Noortwijk 2005; Sobanjo 2011), which vary with time, unlike the stationary transition probabilities used within the conventional Markovian models. Recently, continuous time semi-Markov approaches for modeling bridge deterioration have also been proposed. However, the application has been limited to small illustrative subsets of bridges with mixed results (Ng and Moses 1998; Sobanjo 2011). There is one instance in which deterioration models for a complete state inventory of bridges were developed based on Weibull parametric survival functions (Agrawal et al. 2010). In this study, although survival analysis was used for statistical regression at each condition rating, the subsequent deterioration model was constructed by fitting a polynomial function to the cumulative mean condition rating durations, thus prematurely curtailing the advantages of probabilistic modeling.

The framework presented in this paper develops duration-based probabilistic and multivariable deterioration models combining the advantages of the semiparametric Cox proportional hazards model (PHM) of survival analysis and discrete time nonhomogeneous Markov chain theory. The Cox PHM is used extensively in the medical field for clinical trials, and more recently its applicability in carrying out reliability studies of industrial and infrastructural components has been investigated (Kumar 1995; Mauch and Madanat 2001; Yang et al. 2013). Unlike parametric models like the Weibull function, which cannot model unimodal hazard functions frequently found in infrastructure deterioration (Yang et al. 2013), the proportional hazards approach does not restrict the shape of the hazard function to a particular distribution and has the added advantage of relating the hazard function to various explanatory variables. Owing to its multivariable approach, complete data sets can be used for statistical analysis, thus eliminating the need for categorization and overcoming limitations associated with reduced data sets while enabling improved statistical identification of the covariates most significantly influencing deterioration rates. In this study, PHM survival functions were used to calculate the time-varying probabilities of transition to a lower condition state for all condition states occurring during the life cycle of a bridge component. These transition probabilities across all condition states were inserted into the transition probability matrices at each discrete time interval of a nonhomogeneous Markov chain to stochastically forecast the future condition state of the bridge component. This is unlike the probabilistic Weibull models developed for BMS, which, so far, account for duration dependence in one condition state only (Sobanjo and Thompson 2011). In Weibull-based semi-Markov models proposed in some studies, accounting for more than two condition states becomes computationally complex (Ng and Moses 1998; Sobanjo 2011) and not easily implemented even on small subsets of bridges used for explaining the methodology. Nonhomogeneous Markov chains can be used to probabilistically model the impact of time on the deterioration rate over the entire life cycle in a much simpler format than permitted by the semi-Markov formulation. Moreover, the discrete time formulation of the nonhomogeneous Markov chain is more suited to the discrete nature of condition rating data than a continuous time semi-Markov model and has the added advantage of convenience in implementation within network-level BMSs. Although there are references to applications of nonhomogeneous Markov chain theory in the fields of biostatistics and demography (Fleming and Harrington 1978; Vassiliou 1998), there is no previous reference to its use in bridge deterioration modeling to the authors' knowledge, except as detailed in Goyal (2015).

This paper presents three primary research contributions within a probabilistic deterioration modeling framework. The first part includes a brief overview of the methodology to develop Cox PHM-based survival functions, with a more comprehensive overview in Goyal et al. (2016). The survival functions characterize the duration-dependent impact of significant explanatory factors over the full life cycle of highway bridge components. Additionally, the impact of different censoring protocols on modeling survival functions for deterioration is discussed. The second primary contribution is the derivation of time-dependent transition probabilities of deterioration from the semiparametric PHM survival functions. A novel approach for incorporating the effects of external factors from the multivariable survival functions into the transition probabilities is described. The third primary contribution consists of using the discrete nonhomogeneous Markov chain theory to develop a nonstationary probabilistic approach for bridge deterioration forecasting. Supplemental research contributions include a simplified approach to compute stationary transition probabilities, also with multivariable effects, from the nonstationary PHM transition probabilities, comparative analysis of the developed stationary and nonstationary PHM deterioration models, and analysis of the predictive fidelity of the developed models.

Theoretical Background

Duration models of bridge deterioration are probabilistic models in which the time or duration that a bridge component remains in a particular condition state is treated as a random variable. In addition to time dependence, duration models can also account for incompletely observed or censored observations, frequently found within bridge condition rating duration data. Censored observations include all those instances when the entire duration spent in a particular condition rating has not been recorded. Such instances may occur due to a truncated recording period relative to the duration of the deterioration process or in cases when the natural deterioration process is interrupted by maintenance interventions. Accounting for censoring results in more realistic estimates of deterioration and is the main reason for the superior accuracy of the duration models in comparison with conventional statistical techniques employed by previous deterioration models that cannot account for censoring.

Within survival analysis, if the survival time of a bridge component in a particular condition state is T , its probability density function $f(t)$ provides the probability that $T = t$ at any time t , and its cumulative distribution function $F(t)$ represents the proportion of bridge components with $T < t$ that have already transitioned to a lower rating at time t . The cumulative survival rate $S(t)$, also known as the survival or survivorship function, is the proportion of bridge components expected to stay in the specified condition rating (Lawless 1982; Ng and Moses 1998; Hosmer and Lemeshow 1999; Mauch and Madanat 2001) and is given by

$$S(t) = 1 - F(t) = P(T \geq t) \quad (1)$$

The risk of transitioning to a lower rating at any given time t is measured in terms of the hazard rate function $h(t)$, which is the instantaneous rate of failure, conditional to survival, at time t and is given by

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{P(t \leq T \leq t + \Delta t) | T \geq t}{\Delta t} = -\frac{d}{dt} \ln S(t) \quad (2)$$

The hazard rate is also known as the conditional failure rate and depends on when the observation was made. A constant hazard rate

that does not vary with time implies that the process is memoryless or characterized by duration independence. The hazard rate function may have an upward or a downward slope depending on whether the risk of failure increases or decreases with time. This is termed as positive or negative duration dependence, respectively (Greene 1997).

Cox PHM

The Cox PHM (Cox 1972) is a semiparametric duration model that does not restrict the shape of the hazard function and is also capable of reflecting the effect of various explanatory variables. In the Cox PHM, the hazard rate $h(t, \tilde{z})$ at time t and for covariates $\tilde{z}$ is defined in terms of two components: (1) a nonparametric baseline hazard function, h_0 , which varies only with time, and (2) a time-independent multiplier function using the exponential function to represent the effects of the covariates $\tilde{z}$ through regression coefficients $\tilde{\beta}$, as given by

$$h(t, \tilde{z}) = h_0(t) e^{\tilde{z}\tilde{\beta}} = h_0(t) e^{(z_1\beta_1 + z_2\beta_2 + \dots + z_n\beta_n)} \quad (3)$$

Here, $\tilde{z}$ is a row vector of covariates or explanatory factors and $\tilde{\beta}$ is a column vector of the corresponding regression coefficients that define the effect of the covariates on the hazard rate. The baseline hazard rate $h_0(t)$ is the underlying model for the default factors or those associated with the covariate assignments of zero, which produces $e^{\tilde{z}_0\tilde{\beta}} = 1$.

The multiplier function associated with the nonzero covariates adjusts the hazard rate proportional to the covariates. The relative risk of instantaneous failure of any two items associated with two distinct sets of covariate values is always a constant. The name of the model is derived from this property of proportionality of hazard functions (Kumar and Klefsjö 1994). When the relative risk is evaluated with respect to baseline covariate values, as shown in the following, the resulting value of the proportionality constant is known as the hazard ratio (HR)

$$HR = \frac{h(t, \tilde{z}^1)}{h(t, \tilde{z}^0)} = e^{\tilde{\beta}(\tilde{z}^1 - \tilde{z}^0)} \quad (4)$$

The survival function of an individual subject in a proportional hazards model can be expressed in terms of the covariate vector $\tilde{z}$ for the subject, and the baseline survival function S_0 for $\tilde{z} = 0$, as given by

$$S(t, \tilde{z}) = [S_0(t)]^{e^{(\tilde{z}\tilde{\beta})}} \quad (5)$$

The regression coefficients $\tilde{\beta}$ in the Cox PHM are estimated by maximizing a partial likelihood function derived from the distribution (Cox 1972; Cox and Oakes 1984), suitably modified to handle tied survival times frequently found in applied settings (Kalbfleisch and Prentice 1980). The baseline survival function $S_0(t)$ is subsequently obtained by maximizing the full likelihood function developed on the basis of the previously estimated values of $\tilde{\beta}$ (Lawless 1982). Software programs for proportional hazards regression generally provide an estimator of the baseline survival function in addition to the estimated regression coefficients. Additionally, approaches to handle tied data are also available, the most common being the Breslow approximation (Hosmer and Lemeshow 1999), which is used in the present study.

Discrete analogs of the proportional hazards model also exist (Kalbfleisch and Prentice 1980), including a linear log odds model, commonly known as the Cox proportional odds model (Cox 1972). However, instances of their use in the literature are much less frequent. Conventionally, the continuous Cox PHM has been used in most

applied studies including those specific to infrastructure deterioration modeling (Mauch and Madanat 2001; Samrout et al. 2007; Nakat and Madanat 2008; Yang et al. 2013; Fang et al. 2018). Consistent with prior research, the authors adopted an assumption that the yearly resolution of duration data over a period of 35 years is adequate enough to enable the use of the continuous Cox proportional hazards model in the present study. This assumption is supported by the fully stratified analysis presented in Goyal et al. (2016).

Markovian Probabilistic Approach

A Markov process is defined as a stochastic process having a countable state space, in which the conditional probability P of a future condition state depends only on the present state and is independent of the past states. For a discrete time stochastic process X_t , also called a Markov chain, this can be represented by

$$P(X_{t+1} = i_{t+1} | X_t = i_t, X_{t-1} = i_{t-1}, \dots, X_1 = i_1, X_0 = i_0) = P(X_{t+1} = i_{t+1} | X_t = i_t) \quad (6)$$

where i_t = condition state at time t (Ross 1970). If these conditional probabilities, also known as transition probabilities of the Markov chain, are independent of time t , the process is deemed a stationary or (time) homogeneous Markov chain.

In the context of bridge deterioration, the visual inspection program mandated in the United States results in condition ratings assigned to bridge deck, superstructure, and substructure components on a declining ordinal scale, with 9 representing excellent condition and 0 representing failure. In a Markovian formulation, each condition rating is representative of a possible state of the bridge component being modeled. The change of state is assumed to occur at discrete time intervals equal to a typical reporting period of 1 year. Consequently, the probabilities $P_{i,j}$ that a bridge component would transition from state i to another state j during a specified period can be represented in a transition probability matrix in which the indexes i and j can take any value between the lowest and the highest condition state for the particular bridge inventory database. The transition probability matrix $\mathbf{P}$ is usually represented as

$$\mathbf{P} = \begin{bmatrix} P_{kk} & P_{k(k-1)} & 0 & 0 & \dots & 0 & 0 \\ 0 & P_{(k-1)(k-1)} & P_{(k-1)(k-2)} & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & P_{22} & P_{21} \\ 0 & 0 & 0 & 0 & \dots & 0 & P_{11} \end{bmatrix} \quad (7)$$

where k = highest condition state; and 1 = lowest condition state. Each row of the matrix represents the probability of moving from one state to any other state, including itself. Consequently, the sum of the probabilities in each row should be equal to 1 (Jiang et al. 1988; Morcous et al. 2003). For computational simplicity, it is routinely assumed that a bridge component would not deteriorate by more than one state in a single reporting cycle. This assumption has been found to result in simpler and more robust models (Madanat et al. 1995; Sobanjo and Thompson 2011). The practical influence of these assumptions is reflected in the preceding transition matrix, such that $P_{k(k-1)} = 1 - P_{kk}$, $P_{(k-1)(k-2)} = 1 - P_{(k-1)(k-1)}$, $\dots$, $P_{21} = 1 - P_{22}$, and $P_{11} = 1$. The transition probability of the lowest state P_{11} is 1 because a condition state lower than 1 is usually not recorded in the bridge database, thus

nullifying the possibility of transitioning to a lower state that does not exist. The associated probabilities of each condition rating remaining unchanged between consecutive reporting instances are simply the P_{ii} probability values, which are found on the diagonal of the transition matrix. Furthermore, it is assumed that the deterioration takes place without rehabilitation and the probability of improvement at any state is zero. Therefore, all the probabilities below the diagonal that signify improvement take a value of 0 (Butt et al. 1987; Jiang et al. 1988; Madanat et al. 1995).

The transition probability matrix $\mathbf{P}$ can be used to predict the likelihood of a future condition of a specified bridge component if its present condition is known. Using the conventional Markov chain approach, the condition state of a bridge component at any point in time is represented by a vector of condition rating probabilities; for example, the initial state vector $\mathbf{Z}_0$ for a component in new condition will be [1 0 0 0 0 0 0 0] with a nonzero probability of 1 assigned only to condition rating 9 and a probability of 0 assigned to all other condition ratings from 8 to 1. The future state vector at the end of n years, $\mathbf{Z}_n$, can be predicted using

$$\mathbf{Z}_n = \mathbf{Z}_0 \cdot (\mathbf{P})^n \quad (8)$$

The modified probabilities associated with each condition rating in the resulting future state vector signify the likelihoods that the bridge component will be in each of these ratings after n years. Following the calculation of the future state vector, the expected condition rating at the n th year can be obtained using

$$E = \mathbf{Z}_n \cdot \mathbf{R} \quad (9)$$

where the ordinal values of condition ratings are used to generate the column vector

$$\mathbf{R} = [9 \ 8 \ 7 \ 6 \ 5 \ 4 \ 3 \ 2 \ 1]^T \quad (10)$$

The condition ratings predicted in this way can be used in BMS multiyear optimization iterations to estimate the maintenance, repair, and replacement actions that will likely be required over the planning horizon.

Semi-Markov Approaches

The transition probabilities within the conventional Markovian model described previously are stationary or unchanging with respect to time. Implicit in this model is the unrealistic assumption that the probability of a bridge component deteriorating from its present state to a lower condition state is independent of the duration that it has already spent in the present state. Subsequent to the application of duration modeling approaches to bridge deterioration, some researchers have proposed the use of a semi-Markov approach to take into account the effect of time on the deterioration process (Ng and Moses 1998; Sobanjo and Thompson 2011). The semi-Markov process is similar to a Markov process except that the process is assumed to stay in each state for a random duration, the distribution of which is dependent on the present state and the consecutive state. The process can be seen as a combination of Markov renewal process and Markov chain in which the process renews itself into the same state for a random duration until it transitions or jumps to the next state. It is called semi-Markov because the Markov property of independence from past states is applicable only at the time of the jump to the next state (Cinlar 1975). Semi-Markov processes are generally modeled in terms of continuous time. The simultaneous modeling of two interdependent probabilistic processes, that is, the holding time distribution within a state and the probability of transition to a different state, makes the theory of the semi-Markov process complex and not easily

comprehensible to a nonstatistician (Barbu et al. 2004; Vassiliou and Papadopoulou 1992).

Within semi-Markov processes, the calculation of transition probabilities has been illustrated using continuous time parametric Weibull survival functions (Mishalani and Madanat 2002; Sobanjo 2011). For a survival function $S(t)$ in state k , the probability of dropping to a lower state $k - 1$ within a period Δ after time t is given by

$$P_{k,k-1} = 1 - \frac{S(t + \Delta)}{S(t)} \quad (11)$$

and the probability of remaining in the same state is given by

$$P_{k,k} = \frac{S(t + \Delta)}{S(t)} \quad (12)$$

However, no references were found with regard to the application of transition probabilities calculated in this way to the prediction of bridge deterioration.

Nonhomogeneous Markov Chains

Discussion of Markov probability theory in most textbooks relates only to homogeneous Markov processes (Ross 1970; Cinlar 1975; Osaki 1985). However, the theory and development of nonhomogeneous Markov processes can be tracked through theses and papers over a similar time period (Hajnal 1955; Huang 1977; Fleming and Harrington 1978; Vassiliou 1998). Nonhomogeneous Markov chains are Markov chains as described in Eq. (6) in which the transition probabilities do not remain constant with time t . If $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ are the transition matrices of a nonhomogeneous or nonstationary Markov chain, and the initial state vector of condition rating probabilities is $\mathbf{Z}_0$, the future state vector is obtained using

$$\mathbf{Z}_n = \mathbf{Z}_0 \prod_{i=1}^n \mathbf{P}_i \quad (13)$$

Within nonhomogeneous Markov processes, the application of a discrete time theory in calculation of transition probabilities is found to be more appropriate in cases where the observations are recorded at discrete time intervals (Huang 1977; Fleming and Harrington 1978). Considering the discrete nature of bridge inspection data, this discrete nonstationary Markov chain approach (Goyal 2015) seems more suitable for infrastructure deterioration modeling in comparison with the previously proposed semi-Markov approaches, which are based on the assumption of continuous time.

The following sections describe the theoretical approaches developed during this research for derivation of nonstationary transition probabilities from PHM survival functions and the formulation of a nonhomogeneous Markov chain methodology for deterioration forecasting.

PHM Transition Probabilities with Multivariable Effects

Transition probabilities can be derived from PHM survival functions similar to the approach proposed by earlier researchers for parametric Weibull functions (Mishalani and Madanat 2002; Sobanjo 2011). However, the presence of a defining set of covariates associated with each bridge is an important distinguishing feature of PHM transition probabilities developed in this study. For the derivation of nonstationary PHM transition probabilities in the context of bridge deterioration, let $S_k(t, \tilde{\mathbf{z}})$ be the survival function of a bridge component associated with condition rating k for a bridge described by the vector of covariates $\tilde{\mathbf{z}}$. At any time t , the value of $S_k(t, \tilde{\mathbf{z}})$ is the

cumulative probability that the structural component will remain in condition rating k up to time t . This probability is naturally 1 at $t = 0$ and decreases with each inspection cycle Δ . The survival probabilities so obtained can be used to calculate the transition probabilities $P_{k,k}$. Therefore, the instantaneous probability that the structural component will remain at the same condition rating k over the next annual reporting cycle at any time t is given by

$$P_{k,k}(t, \tilde{\mathbf{z}}) = \frac{S_k(t + \Delta, \tilde{\mathbf{z}})}{S_k(t, \tilde{\mathbf{z}})} = \frac{S_k(t + 1, \tilde{\mathbf{z}})}{S_k(t, \tilde{\mathbf{z}})} \quad \text{for } \Delta = 1 \text{ year} \quad (14)$$

Because it is assumed that improvement in condition rating is not possible in the presence of deterioration without rehabilitation, the probability of deteriorating to condition rating $(k - 1)$ at any time t is consequently given by

$$P_{k,k-1}(t, \tilde{\mathbf{z}}) = 1 - P_{k,k}(t, \tilde{\mathbf{z}}) \quad (15)$$

The preceding transition probabilities do not yet incorporate the influence of covariates included in the PHM survival functions on the hazard rate. A novel method was developed in this study to use the basic output of a proportional hazards model, which is composed of the baseline survival function and the PHM regression coefficients, to calculate unique transition probabilities for any individual bridge component, or category of bridge components, associated with a specific set of covariate values. If $\mathbf{z}^0$ denotes the set of baseline covariates, $S_k(t, \mathbf{z}^0)$ can be defined as the baseline survival function and $P_{kk}(t, \mathbf{z}^0)$ can be defined as the baseline stay-the-same transition probability for condition rating k . For any other set of covariates, denoted as $\mathbf{z}^1$, associated with assignments in a multivariable model, PHM allows for calculation of a hazard ratio [refer to Eq. (4)]

$$\text{HR}_k = e^{\tilde{\beta}_k(\mathbf{z}^1 - \mathbf{z}^0)} = e^{\beta_{k,1}(z_1^1 - z_1^0) + \beta_{k,2}(z_2^1 - z_2^0) + \dots + \beta_{k,n}(z_n^1 - z_n^0)} \quad (16)$$

such that [refer to Eq. (5)] the covariate-specific survival function $S_k(t, \mathbf{z}^1)$ is given by

$$S_k(t, \mathbf{z}^1) = S_k(t, \mathbf{z}^0)^{\text{HR}_k} \quad (17)$$

Therefore, using Eq. (14)

$$\begin{aligned} P_{k,k}(t, \mathbf{z}^1) &= \frac{S_k(t + \Delta, \mathbf{z}^1)}{S_k(t, \mathbf{z}^1)} = \left[\frac{S_k(t + \Delta, \mathbf{z}^0)}{S_k(t, \mathbf{z}^0)} \right]^{\text{HR}_k} \\ &= [P_{k,k}(t, \mathbf{z}^0)]^{\text{HR}_k} \end{aligned} \quad (18)$$

and

$$P_{k,k-1}(t, \mathbf{z}^1) = 1 - [P_{k,k}(t, \mathbf{z}^0)]^{\text{HR}_k} = 1 - P_{k,k}(t, \mathbf{z}^1) \quad (19)$$

where $P_{k,k}(t, \mathbf{z}^1)$ and $P_{k,k-1}(t, \mathbf{z}^1)$ are the transition probabilities associated with the $\mathbf{z}^1$ set of covariates. Transition probabilities calculated in this way capture the multivariable effects of the PHM survival function in addition to the effect of duration dependence. In the context of computational resource requirements, it may be noted that although the basic output of a proportional hazards model is composed of the baseline survival function and the PHM regression coefficients, the survival functions are not directly used beyond developing the baseline transition probabilities. Therefore, for practical implementation, the BMS needs only to store baseline transition probabilities for each model and the corresponding hazard ratios associated with the specific bridge covariate assignments over each condition rating. These are small matrices with nominal storage requirements and insignificant computational resources are needed to carry out the subsequent construction of bridge-specific transition probability matrices. The use of these transition probabilities in a Markovian formulation

for bridge prediction carries this advantage forward to the deterioration forecasting process as explained in the following section.

Nonstationary Markov Chain Forecasting of Bridge Deterioration

A simple formulation was developed in this study for using the nonhomogeneous Markov chain theory in modeling bridge

deterioration. For any bridge component, time-dependent and covariate-specific transition probabilities computed in accordance with the method detailed in the previous section are first assembled to develop transition probability matrices for each year of the planning horizon. For example, considering a planning horizon of n years, if the duration of each prediction cycle Δ is 1 year, the transition probability matrix for the i th year, where $i = 1, 2, \dots, n$, is developed using

$$\mathbf{P}_i = \begin{bmatrix} P_{m,m}(i, \tilde{\mathbf{z}}) & P_{m,m-1}(i, \tilde{\mathbf{z}}) & 0 & 0 & \dots & 0 & 0 \\ 0 & P_{m-1,m-1}(i, \tilde{\mathbf{z}}) & P_{m-1,m-2}(i, \tilde{\mathbf{z}}) & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & P_{2,2}(i, \tilde{\mathbf{z}}) & P_{2,1}(i, \tilde{\mathbf{z}}) \\ 0 & 0 & 0 & 0 & \dots & 0 & P_{1,1}(i, \tilde{\mathbf{z}}) \end{bmatrix} \quad (20)$$

In this matrix, m is the highest condition state and 1 is the lowest condition state. A simplifying assumption made in the development of nonstationary transition probability matrices is that the time spent in the initial condition state is 0 (Sobanjo 2011). The transition matrix of any year contains the probability of staying in the same condition state, as well as the probability of deteriorating to the next lower condition state in that year, for each condition state in which a bridge component may be. Now, if the initial state vector $\mathbf{Z}_0$ for a component is known, the predicted state of condition rating probabilities at year n , $\mathbf{Z}_n$, can be obtained by applying Eq. (13) recursively as follows:

$$\mathbf{Z}_1 = \mathbf{Z}_0 \cdot \mathbf{P}_1 \text{ after first year} \quad (21)$$

$$\mathbf{Z}_2 = \mathbf{Z}_1 \cdot \mathbf{P}_2 \text{ after second year} \quad (22)$$

$$\mathbf{Z}_3 = \mathbf{Z}_2 \cdot \mathbf{P}_3 \text{ after third year} \quad (23)$$

where $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3, \dots$ are the predicted state vectors at the end of the first year, second year, third year, and so on until the n th year prediction is obtained as

$$\mathbf{Z}_n = \mathbf{Z}_{n-1} \cdot \mathbf{P}_n \text{ after } n\text{th year} \quad (24)$$

Implemented this way, the algorithm produces a vector output at each year, thus considerably simplifying the matrix multiplication process envisaged in Eq. (13). Moreover, because the transition probabilities of the nonhomogeneous Markov chain are derived from the survival probability distributions of individual condition states, it also accounts for the aspect of duration dependence in the deterioration prediction process.

PHM Bridge Deterioration Model and Its Implementation

The PHM bridge deterioration model is composed of the baseline transition probabilities and covariate-specific hazard ratios associated with all condition ratings. The framework for arriving at the PHM bridge deterioration model is presented schematically in

Fig. 1. The model was implemented on the North Carolina state bridge inventory, which has more than 17,000 in-service bridges and an inspection database with more than 35 years of digitally recorded National Bridge Inventory (NBI) general condition ratings. The implementation encompassed data from all deck, superstructure, and substructure condition ratings from the years 1981–2015, of which the results obtained from only the subset comprising concrete decks are presented here for illustration. The main emphasis of this paper is on the development of transition probabilities and deterioration prediction protocol subsequent to the development of the PHM survival functions. However, the steps leading up to the survival functions are also described here briefly with reference to earlier publications. The interested reader is referred to these prior publications for more extensive details on the data preprocessing and Cox proportional hazards statistical regression for the development of the survival functions (Goyal 2015; Goyal et al. 2016).

Development of PHM Survival Functions

Relevant descriptive and condition-specific data for each bridge were queried and extracted from external source files comprising the bridge database to construct a set of bridge records for the state inventory. In the cases where a bridge was reconstructed or rebuilt, the bridge record was split into two separate records so the record for the period after the reconstruction or rebuilding was distinct from the earlier record. These data were further preprocessed to extract all observations of the response variable, which is the observed continuous duration at the particular condition rating being analyzed. For this purpose, only observations with a duration of at least 2 years were considered due to the biennial frequency of inspection. The descriptive information on each structure, such as its functional classification, geographical location, traffic characteristics, design parameters, and other details contained within the BMS historical records that could potentially influence the deterioration rates of specific bridge components were considered as potential explanatory factors. Each of these factors was organized into categories designated by one or more design variables to which bridges were classified based on either binary or reference cell coding. The variables of average daily traffic (ADT), average daily

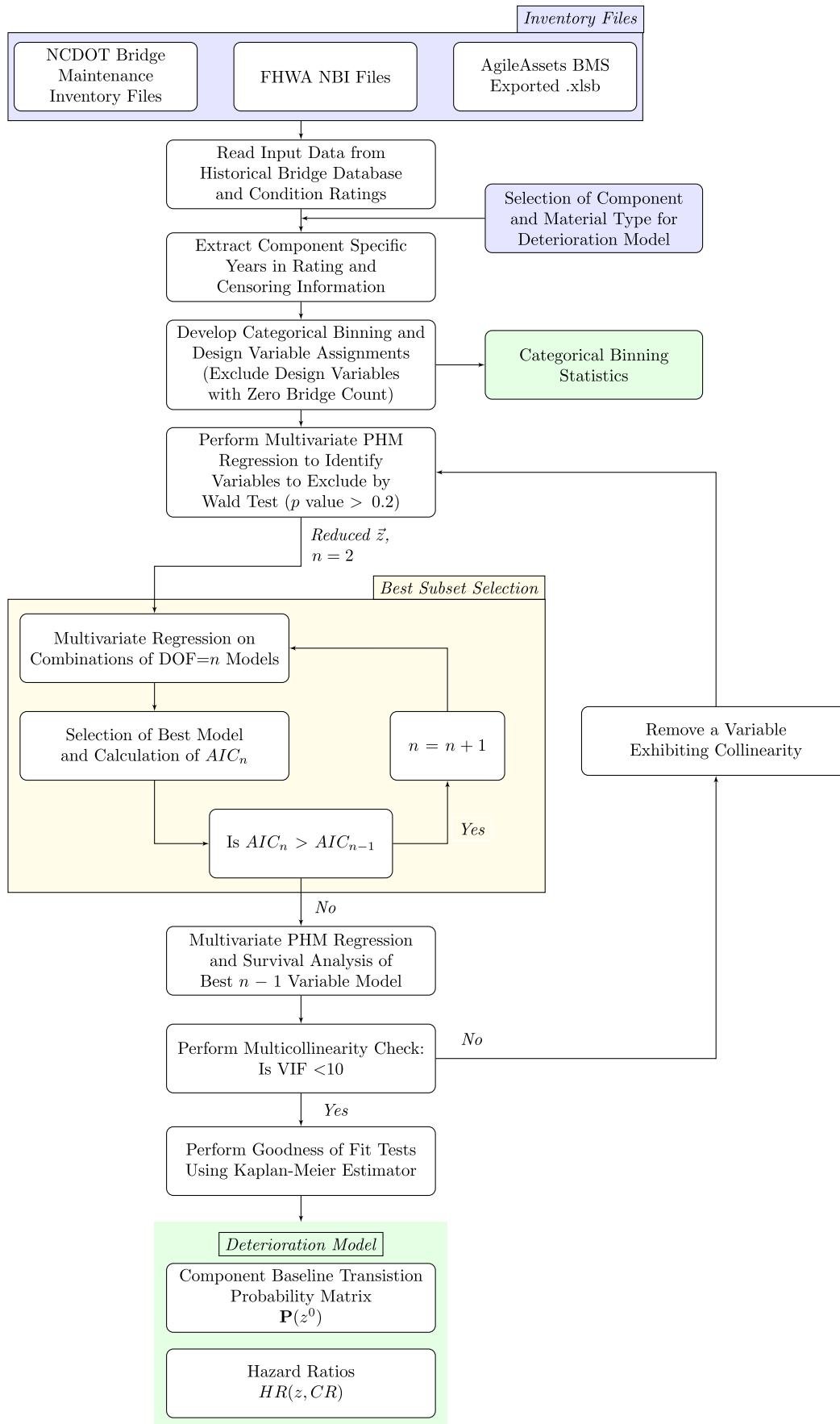


Fig. 1. PHM bridge deterioration model development flowchart.

Table 1. Categorical covariates included in PHM analysis for concrete deck deterioration model

Covariate	Design variable
State system	Primary/interstate ^a Secondary
Reconstruction	Original or rebuilt ^a Reconstructed
Region	Coastal ^a Piedmont Mountain
Wearing surface	None ^a Monolithic concrete Integral concrete Latex concrete Low-slump concrete Bituminous Timber Gravel Other
Number of spans	Single ^a Multiple
Average daily traffic	$0 \leq \text{ADT}^a < 878$ $878 \leq \text{ADT} < 3,184$ $3,184 \leq \text{ADT} < 9,090$ $\text{ADT} \geq 9,090$
Average daily truck traffic	$0 \leq \text{ADTT}^a < 57$ $57 \leq \text{ADTT} < 226$ $226 \leq \text{ADTT} < 941$ $\text{ADTT} \geq 941$
Maximum span (m)	$0 \leq \text{MaxSpan}^a < 4$ $4 \leq \text{MaxSpan} < 6$ $\text{MaxSpan} \geq 6$
Age (years)	$0 \leq \text{Age}^a < 14$ $14 \leq \text{Age} < 23$ $23 \leq \text{Age} < 33$ $\text{Age} \geq 33$

^aBaseline or reference category.

truck traffic (ADTT), age, and maximum span length, which are continuously recorded, were divided into categories of approximately equal frequency of occurrence for each component based on weighted averages computed across the available bridge records. Therefore, the categorical ranges for these variables differed for different component and material types depending on the statistical distribution (Goyal et al. 2016). The category bounds developed for concrete deck components are provided in Table 1 for reference. Distributions of observed continuous duration records by bridge age for condition ratings 9 through 4 are shown in Fig. 2 with reference to the categories associated with the covariate age in Table 1. Age refers to the time the deck enters the observed condition state (or the beginning year of the duration record) from when the bridge was last constructed or reconstructed. Although the covariates used in this study were practically limited to those available within BMS and related databases, it is natural that the models would have benefited from inclusion of more likely factors associated with deterioration, for example, slab thickness, concrete cover, chloride permeability, and climatic factors like temperature and precipitation. There is a broader need within BMS practice to reconsider and expand the set of deterioration-related data items stored in BMS databases with a view toward facilitating improved bridge deterioration models.

In addition to the independent and dependent variables, censoring information was compiled in a separate vector of the same size as the response variable as a binary variable of either 0 or 1 depending on whether the observations were classified as either fully

observed or censored. The distribution of condition rating duration records and their censoring characteristics for concrete bridge decks in the state of North Carolina are summarized in Table 2. It can be observed that very few uncensored records, if any, are available for condition ratings 1, 2, and 3. This is because very few bridge components are permitted to deteriorate below condition rating 3. Consequently, the development of survival functions was limited to condition ratings 9 through 4.

The censoring protocol adopted in this study and followed in this paper classifies as censored all condition rating durations that were not observed completely at the beginning and end years of the database. In addition, instances of improvement in condition rating at the end of the duration were considered censored under the assumption that maintenance and repair actions prematurely shortened the duration in the current condition rating under natural deterioration. This censoring protocol produces flatter survival functions for the lower condition ratings in comparison to survival functions obtained for the higher condition ratings. Interaction with North Carolina DOT (NCDOT) engineers indicated that most repair and maintenance actions were triggered by a jump to a lower rating that was spotted during inspection, and therefore the improvement in condition rating subsequent to the maintenance action was often reflected in the inspection record before any sign of transition to the lower rating. The authors discovered that modifying the censoring protocol to classify all instances of censoring due to elevated condition ratings as not censored, based on this presumption, increased the slope of the survival functions at ratings 7 and lower. However, modifying the censoring protocol in this way also resulted in greater prediction errors at these ratings, with the effect being most pronounced at condition rating 4. At this point, it is important to understand that routine maintenance actions are considered an inseparable part of natural deterioration for the purposes of duration analysis because there is no simple way to separate out the effect of routine actions based on the available maintenance records. Consequently, it is logical to assume that because the frequency of routine maintenance actions increases significantly at lower ratings, it is likely to give incorrect results if not accounted for. It was concluded by the authors that censoring at intermediate data points for instances of elevated condition rating accounts for the effect of increased maintenance frequency giving a much improved prediction fidelity than obtained otherwise. Fig. 3 presents the baseline survival functions obtained from PHM regression of the concrete bridge deck data set. The survival functions produced by excluding the maintenance-related censoring at intermediate data points are also presented with dashed lines for comparison.

The steps in the modeling process flowchart involving multi-variable PHM regression, including the best subset selection and the development of survival functions, are performed individually on each of the condition-rating-specific sets associated with the component subset analyzed. PHM regression was carried out initially using only those design variables that were observed within one or more bridge records for the condition rating under study. From this initial regression, the variables that were found to be statistically significant with a Wald statistic p -value of less than or equal to 0.2 were included in the best subset selection routine. Best subset selection is essentially a strategy for optimally reducing the size of the model without compromising its reliability. For this study, the best subset selection routine was designed as a constrained stepwise forward selection. The best subset selection routine began with two variables and then the number of variables was incrementally increased by selecting the combination of variables at each level that maximized the log partial likelihood. The Akaike information criterion (AIC) was calculated for all combinations of

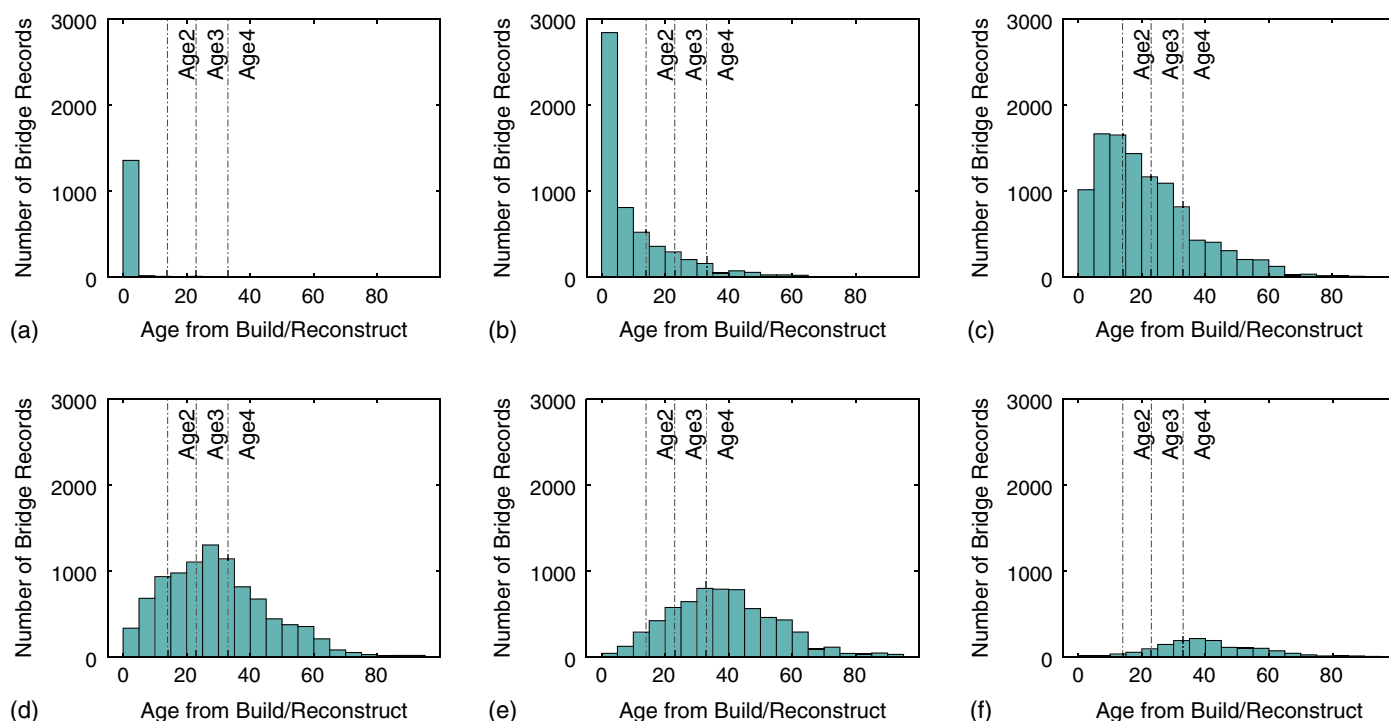


Fig. 2. Distributions by bridge age at the beginning of observed continuous duration records of concrete decks at condition ratings 9 through 4 (dashed lines represent lower bounds of the PHM categorical variable age): (a) condition rating 9; (b) condition rating 8; (c) condition rating 7; (d) condition rating 6; (e) condition rating 5; and (f) condition rating 4.

Table 2. Concrete deck condition rating data overview

Deck condition	NBI rating	Total records	Uncensored records	Censored records	Percentage of censored records
Excellent	9	1,419	352	1,067	75.2
Very good	8	5,477	2,696	2,781	50.8
Good	7	10,544	3,389	7,155	67.9
Satisfactory	6	9,541	4,051	5,490	57.5
Fair	5	6,555	885	5,670	86.5
Poor	4	1,452	95	1,357	93.5
Serious	3	201	2	199	99.0
Critical	2	2	0	2	100.0
Imminent failure	1	4	0	4	100.0

variables included in the model. The number and combination of variables that maximized AIC were taken to constitute the best subset of statistically significant covariates, which was then included in a multivariable PHM regression to estimate the regression coefficients, hazard ratios, and baseline survival function associated with that rating. This best-subset model was tested for potential multicollinearity issues using the variance inflation factor (VIF) indicator. The goodness of fit of the final model was also assessed by developing empirical Kaplan-Meier estimators on the select categorical data at each rating, which confirmed the appropriateness of the proportional hazards assumption (Goyal 2015; Goyal et al. 2016). The survival function developed using this best-subset model incorporates the effect of the most significant explanatory variables on the deterioration rate over an individual condition rating. The state-dependent best subsets and associated hazard ratios obtained for the significant explanatory factors identified in the concrete deck model are summarized in Table 3. An HR value of 1 signifies a lack of influence on the deterioration rate, a value less than 1 indicates that the factor is associated with a reduced rate

of deterioration, and a value greater than 1 indicates that the factor is associated with an increased rate of deterioration. The outcome of lack of influence of a variable on the deterioration rate represented by an HR value of 1 could also be a consequence of very few bridges associated with that variable. For example, in Table 3, the HR value of 1 associated with older age categories at condition rating 9 can be attributed to the fact that very few bridge decks over the age of 23 years are likely to be rated at condition rating 9 (Fig. 2). At the opposite end of the spectrum, the HR values associated with the younger age categories at condition ratings 5 and 4 can similarly be attributed to limited data for selected combinations of age and condition rating because few bridge decks younger than 23 years may be expected to deteriorate to condition rating 4 or 5. It can be observed that the relationship between age and hazard ratio follows an expected trend in all instances of condition ratings such as 6, 7, and 8, where all age categories were found to be significant. A detailed discussion of the state-dependent influence of explanatory factors on deterioration rates based on implementation of the developed PHM bridge deterioration model on all deck, superstructure, and substructure components, including concrete bridge decks, was presented in Goyal et al. (2016).

Development of Nonstationary Transition Probabilities

The semiparametric nature of the PHM survival function complicates the calculation of the baseline transition probabilities associated with staying in the same condition rating at the end of each annual prediction cycle previously presented in Eq. (14). Principally, the PHM survival functions (Fig. 3) exhibit a stepwise nature unlike the continuous and differentiable nature of parametric survival functions. Although each interval of survival time is associated with a unique survival probability in a PHM survival function, the instantaneous drop in the survival probability at the end of each interval to the lower step associated with the subsequent survival

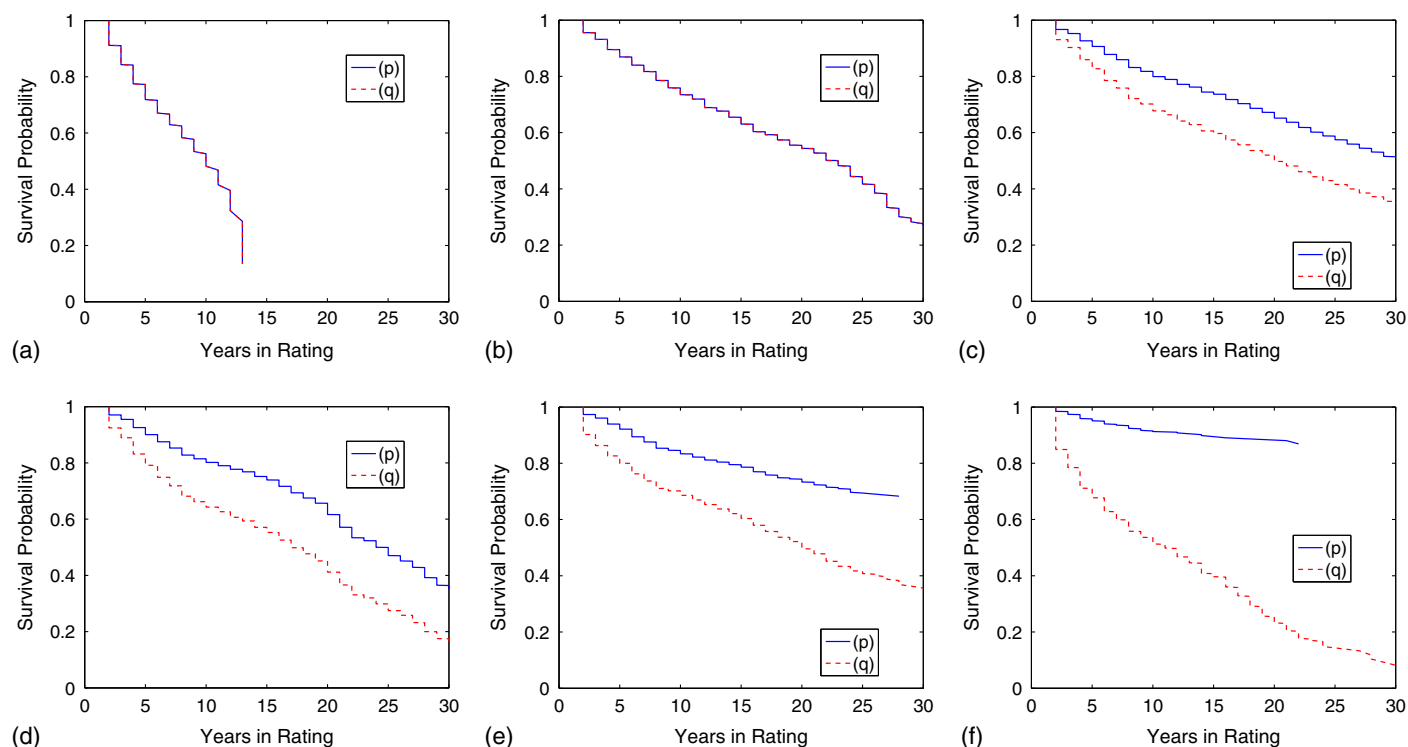


Fig. 3. Baseline survival functions for concrete deck across condition ratings 9 through 4, where (p) includes maintenance-related censoring and (q) excludes maintenance-related censoring: (a) condition rating 9; (b) condition rating 8; (c) condition rating 7; (d) condition rating 6; (e) condition rating 5; and (f) condition rating 4.

Table 3. PHM hazard ratios obtained from concrete deck best subset selection

Factor	Rating					
	9	8	7	6	5	4
State system	1	1	1	1.104	0.790	1
Reconstruction	1	1	1	1.146	1	1
Piedmont	1	1	0.917	0.766	1.423	1
Mountain	1	1	0.697	0.818	1	1
Monolithic	1	1	1.322	1	1	1
ADT4	1	1	1	1	1	1.592
ADTT4	1	1	1	1.180	1	1
MaxSpan2	1	1.509	0.854	1	1	1
MaxSpan3	1	2.107	1	1.389	1	1
Number of spans	1	1.178	1.442	1.323	1	1
Age2	1.980	1.467	1.142	1.237	1	0.253
Age3	1	1.936	1.467	1.446	1.325	1
Age4	1	1.929	2.393	2.250	1	1

interval generates a discontinuity in the survival probabilities at that instant when one interval ends and the next one begins. Also, the survival interval represented by one step is not consistently equal to one inspection interval but sometimes spans across multiple or partial inspection intervals. Therefore, a special subroutine was designed to extract unique values of $S_k(t, \bar{z})$ for each inspection interval of 1 year. The maximum survival probability recorded at any year was assigned to that year. The problem of intervals spanning multiple years was resolved by linear interpolation of the survival probabilities associated with the years at the beginning and end of such an interval.

Baseline transition probabilities calculated in this way from the survival functions in Fig. 3 are presented in Fig. 4. It can be observed that the stay-the-same transition probabilities for each

of the condition ratings 8–4 are nearly constant with the duration that the deck has been continuously rated in the respective condition rating, with some gradual decrease in the stay-the-same transition probability after approximately 15–20 years in that condition rating. In the case of condition rating 9, greater variability is observed with a mild decrease in the stay-the-same transition probability at earlier than 10 years. The visible increase in stay-the-same transition probabilities in the odd numbered years at each condition rating is a reflection of the biennial frequency of the typical inspection cycle.

PHM-Based Deterioration Forecasting

For the purpose of deterioration forecasting, the PHM transition probabilities at all condition ratings are grouped in a single square transition probability matrix as explained previously. In developing practical transition probability matrices for implementation in the BMS, it is important to recognize that very few bridges are permitted to deteriorate below condition rating 3 as reflected in Table 2. For this reason, in developing the earliest Markovian bridge deterioration models (Jiang et al. 1988), the transition probabilities for condition ratings 1 and 2 were excluded from the transition probability matrix, and condition rating 3 was considered the lowest possible or absorbing state. In the current study, it was possible to develop transition probabilities through survival analysis for all the condition ratings from 9 to 4 due to the availability of sufficient historical condition rating data. However, instead of considering the next lower rating of 3 as the absorbing state, the stay-the-same transition probabilities for condition ratings 3 and 2 have been prescribed as 0.85 in the models developed in this study to prevent the deterioration models from converging abruptly to condition rating 3. Condition rating 1 has been considered the lowest possible state with a stay-the-same transition probability of 1.

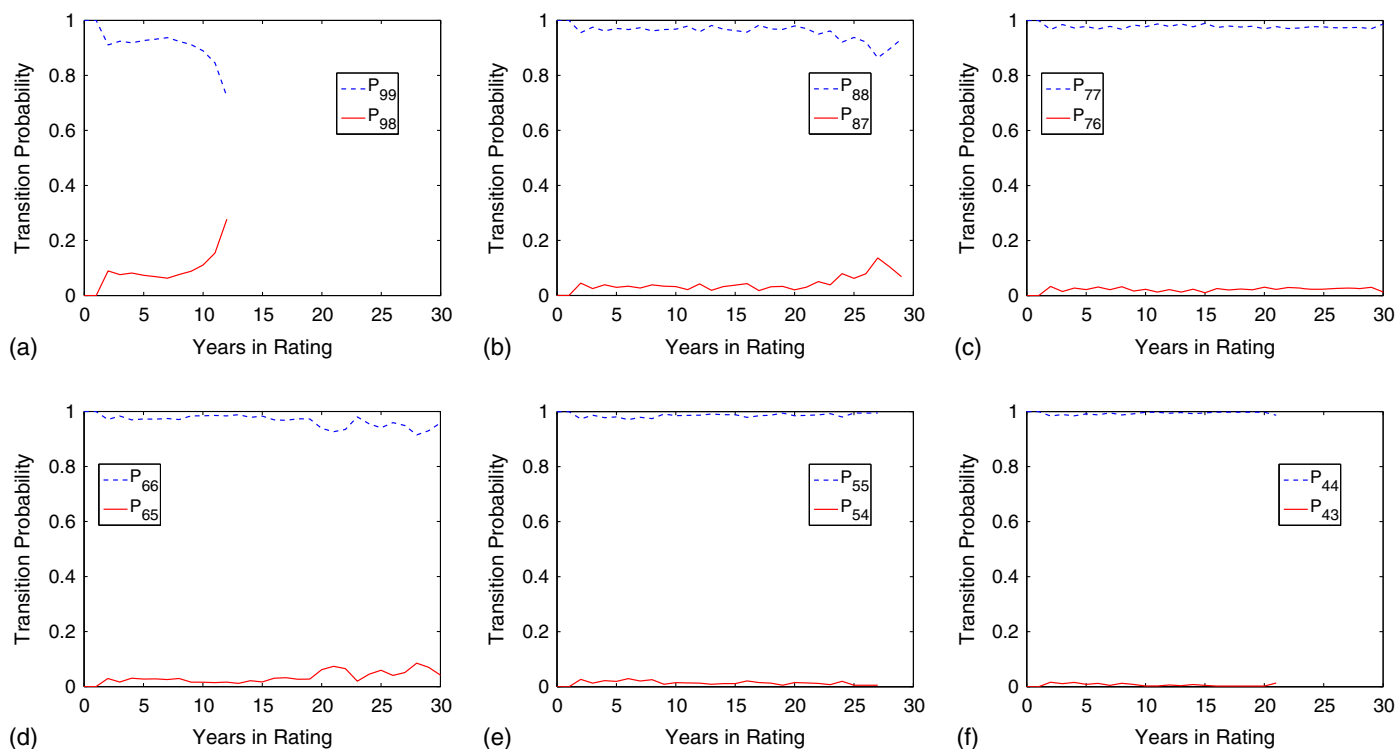


Fig. 4. Baseline transition probabilities for concrete deck across condition ratings 9 through 4: (a) condition rating 9; (b) condition rating 8; (c) condition rating 7; (d) condition rating 6; (e) condition rating 5; and (f) condition rating 4.

For a nonhomogeneous Markov chain prediction protocol, transition probability matrices can be developed for each year in the planning horizon using Eq. (20) based on the values of the transition probabilities in that year. As explained previously, observed transition probabilities can be calculated over the entire duration of the survival function, which is less than or equal to the period of observation. The NCDOT database employed in this study contains observed inspection data for more than 35 years, which allowed for calculation of observed transition probabilities over an extended period. Beyond this observed duration, the transition probability has been assumed to remain constant for the remaining prediction period. In practical implementation, the planning horizon in long-term analyses does not typically exceed 10 or 20 years. However, longer horizons are used to describe the characteristics of the deterioration model over the full service life of the component. These matrices can then be used to predict the future condition of a bridge component using the Cox PHM deterioration model if its vector of descriptive covariate assignments and present condition are known.

There is no reference in the literature to the use of stationary transition probabilities based on survival analysis. However, the general uniformity observed in the transition probabilities developed from PHM survival functions for the bridge deck, superstructure, and substructure components in this study prompted the idea of averaging the transition probabilities over the condition rating duration. Therefore, a stationary transition probability using the mean value of the transition probabilities calculated over the duration of the survival function is also proposed in this study. The deterioration forecasting in this case can proceed similar to the conventional Markov chain models using Eq. (8). A comparative analysis of condition rating predictions obtained using both the stationary and nonstationary formulations for concrete decks is presented subsequently.

A stationary transition probability matrix can also be obtained by using the stationary transition probabilities for individual condition ratings. The stationary transition matrix $\mathbf{P}$ obtained in this way for the concrete decks in the North Carolina bridge inventory is given as follows:

$$\mathbf{P} = \begin{bmatrix} 0.9034 & 0.0966 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.9570 & 0.0430 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.9780 & 0.0220 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.9662 & 0.0338 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.9860 & 0.0140 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.9933 & 0.0067 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.85 & 0.15 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.85 & 0.15 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (25)$$

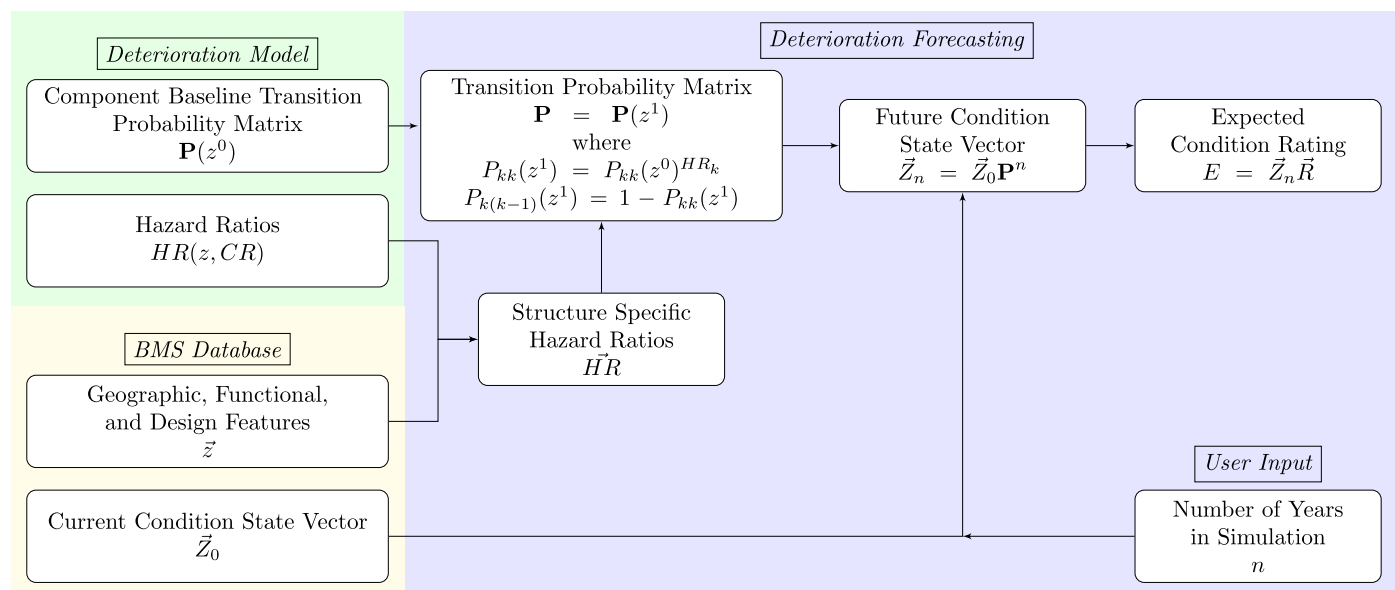


Fig. 5. PHM bridge deterioration model prediction flowchart.

The PHM bridge deterioration model prediction flowchart in Fig. 5 summarizes the progression of PHM-based deterioration forecasting. In the flowchart, the transition probability matrix $\mathbf{P}$ refers to a stationary PHM-based transition probability matrix. For nonstationary prediction, it would be replaced by $(\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n)$, representing the transition probability matrices for each year within the prediction period. In that case, the stationary prediction equation for calculating the future condition state vector $\mathbf{Z}_n$ would also be replaced with the nonstationary formulation given in Eq. (13).

The condition rating predictions obtained for North Carolina concrete bridge decks associated with baseline covariate values using both the nonstationary and stationary prediction protocols are presented in Fig. 6. It can be observed that the deterioration curves obtained from both the stationary and nonstationary formulations exhibit a consistently decreasing trend, which is to be expected. However, this consistency is particularly significant because it could not be achieved in predictions reported in earlier studies using the Markovian probabilistic approach. These previous Markovian approaches include both the stationary Markov chain models in which linear-regression-based methods were employed to calculate the transition probabilities as well as nonstationary semi-Markovian models. Among the stationary Markov chain models, the transition probabilities have been found to approach unity, resulting in an unnatural flattening of the deterioration curve in one study using the New York State DOT bridge inventory (Agrawal et al. 2009, 2010). A secondary Markov process had to be implemented in this study to eliminate this effect. A similar effect was observed in two other studies in which unreasonably high median transition times, sometimes exceeding 100 years, were obtained for prestressed concrete superstructure elements using stationary transition matrices for the bridge inventories in Florida (Sobanjo and Thompson 2011) and Colorado (Hearn 2012). In another study proposing a semi-Markov formulation for prediction of expected condition ratings using nonstationary transition probabilities, a flattening of the deterioration curve was also observed after about 15 years (Sobanjo 2011). The PHMs developed in the current study are able to predict deterioration realistically for more than 100 years using both the traditional stationary Markov chain approach as well as the newly developed nonstationary formulation.

Nonstationary versus Stationary Forecasting

A comparative analysis of condition rating predictions obtained using both the stationary and nonstationary formulations reveals that deterioration rates of all bridge deck, superstructure, and substructure components obtained from both approaches are in close agreement over the first 20–30 years of the prediction period. In the case of concrete decks, as observed in Fig. 6, this agreement extends to nearly 50 years, and moreover, the nonstationary model provides a more optimistic prediction than the stationary model over the long-term projection. The reason for this, as can be observed in Fig. 4, is the variation in concrete deck baseline transition probabilities over the condition rating duration, which are nearly constant in ratings 4 through 8, but the stationary model is biased by the lower mean transition probability in condition rating 9, as

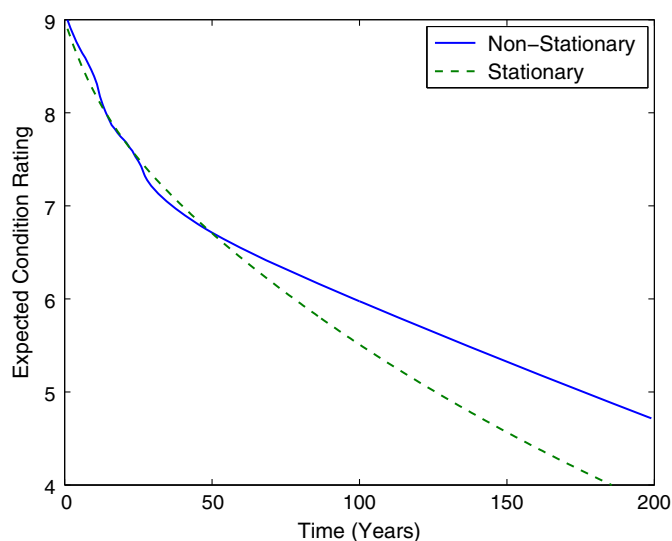


Fig. 6. Expected values of condition rating for concrete deck determined using stationary and nonstationary baseline transition probabilities.

reflected in Eq. (25). Therefore, use of nonstationary transition probability matrices is recommended for greater accuracy in predictions because it accounts for the variation in transition probabilities, especially for long-term planning periods exceeding 20 years.

PHM Multivariable Advantage

Deterioration models can be obtained for bridge components featuring covariate assignments other than the baseline values by first calculating the structure-specific HR for each condition rating using Eq. (16), and then using these HR values to scale the baseline transition probabilities using Eqs. (18) and (19), as explained subsequently. Let $HR_9, HR_8, HR_7, \dots, HR_4$ be the structure-specific hazard ratios describing the effect of associated covariates. Then, for concrete decks at condition rating 8, HR_8 can be calculated

using the HR values from Table 3 if the covariates associated with a particular bridge are known. For example, with reference to Table 1, if a bridge is on State System 2, has not been reconstructed, is located in the mountain region, has no additional wearing surface, has an ADT of 200 vehicles and an ADTT of 20 vehicles, a maximum span length of 5 m, is a single-span structure, and is 25 years old, HR_8 will be the product of the HRs associated with the covariates taking assignments other than the designated baseline values. In this case, these covariates are State System, Mountain, MaxSpan2, and Age3, giving $HR_8 = 1 \times 1 \times 1.509 \times 1.936 = 2.921$. The HRs associated with the other condition ratings can be similarly calculated. Now if $P_{99}, P_{88}, P_{77}, \dots, P_{44}$ are the baseline stay-the-same transition probabilities (stationary or nonstationary), the transition probability matrix that accounts for the effects of design, geographic, and functional features takes the form

$$P = \begin{bmatrix} P_{99}^{(HR_9)} & 1 - P_{99}^{(HR_9)} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & P_{88}^{(HR_8)} & 1 - P_{88}^{(HR_8)} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & P_{77}^{(HR_7)} & 1 - P_{77}^{(HR_7)} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & P_{66}^{(HR_6)} & 1 - P_{66}^{(HR_6)} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & P_{55}^{(HR_5)} & 1 - P_{55}^{(HR_5)} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & P_{44}^{(HR_4)} & 1 - P_{44}^{(HR_4)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.85 & 0.15 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.85 & 0.15 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (26)$$

If we substitute the values of the stay-the-same transition probabilities from the baseline transition probability matrix in Eq. (25), along with the structure-specific HRs calculated previously, we will obtain the structure-specific transition probability matrix for the concrete deck of the bridge in our example.

A unique advantage of PHM is that the hazard ratio of a covariate in a multivariable model still quantifies the individual effect of the specific covariate on the hazard rate when the values of all other covariates in the model are held constant. The construction of transition probability matrices and implementation of the Markovian prediction protocol integrates these effects from individual condition rating survival functions into a multivariable PHM deterioration model spanning the entire life cycle of the component. Fig. 7 illustrates the individual as well as combined effects of two variables, number of spans and age, on concrete deck deterioration rates. The figure illustrates the depth of information revealed by the models developed in this study based on statistically identified significant factors affecting concrete deck deterioration, in contrast to the judgment-based a priori grouping used in present-day BMSs. A case in point are the legacy deterministic deterioration models for concrete bridge decks in the NCDOT BMS that have been constructed using ADT as the only preclassifier. Very insignificant differentiation was observed between these models for different ADT categories, which is explained by the fact that ADT has not been identified as a significant variable affecting concrete deck deterioration in the PHM analysis (Goyal 2015).

Predictive Fidelity of the Developed Models

A significant constituent of the applied contribution of this research was the quantitative assessment of the predictive fidelity of the developed probabilistic models. Ideally, model assessment should be performed using data independent from the records used to develop the statistical models. However, due to the rate of collection of bridge condition rating data, assessment of the predictive fidelity of the developed deterioration models to future response data will not be possible for several years. Consequently, model assessment was performed using a selected time period of condition rating data extracted from the existing NCDOT historical bridge management database. For the assessment results presented here, the 15-year period from 2000 to 2015 was selected. The individual condition rating databases were parsed to obtain only those records that were continuously observed between the starting year and the ending year of the observation period. Additionally, to minimize the presence of observations with significant maintenance or reconstruction actions, the data were filtered to exclude those records where the observed condition rating increased by more than one condition rating between consecutive years. The objective of this data preprocessing was to reduce the data set used for model assessment to only those structures exhibiting typical deterioration under either natural conditions or routine maintenance actions while also accounting for the effects of inspector subjectivity or recording anomalies in the condition rating database. For each of the continuously observed condition rating records satisfying the filtering

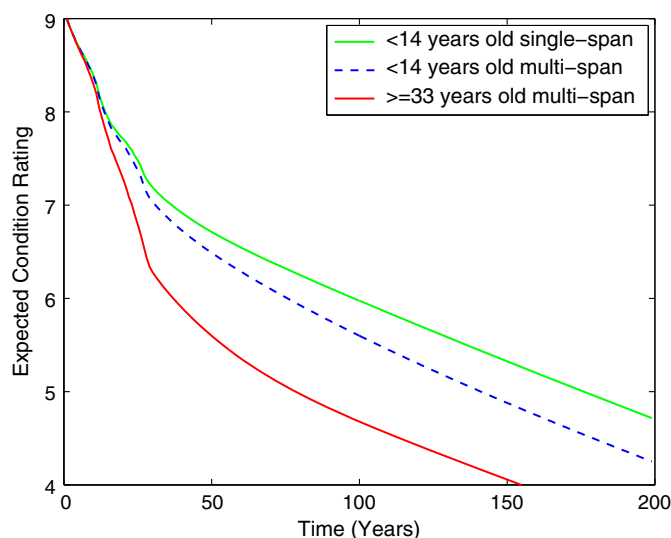


Fig. 7. Effect of age and number of spans on deterioration rates of concrete deck bridges throughout the life cycle.

criteria, covariate assignments for each record were developed from the descriptive data in the bridge record in accordance with the classification scheme of the model applicable to each individual bridge component. Further, in addition to the current rating at the start of the observation period, the duration already spent by the bridge

component at that initial condition rating was also determined and the sum of this initial duration and the 15-year period of the planning horizon was used to obtain expected condition ratings at the end of the prediction period (2015) for each individual bridge record. The actual observed condition rating at the end of the 15-year observation period was subtracted from the predicted condition rating obtained from each model to compute the prediction errors associated with the application of each model to all selected records.

The results from applying the concrete deck deterioration models are presented here for the purpose of illustration. However, the results and conclusions drawn from model assessment using data from other bridge components and material types were found to be strongly consistent with those reflected in the concrete deck analysis. A total of 6,017 bridge records for concrete decks remained after the prefiltering algorithm was applied to the extracted 2000–2015 condition rating data. Statistics of these data, including the initial deck condition rating in 2000, the change in condition rating over the 15-year period, and the preexisting durations that the decks had already experienced at the initial condition ratings, are presented in Fig. 8. Fig. 9 presents the histograms of prediction error distributions for the PHM stationary and nonstationary probabilistic models along with the mean errors and standard deviations. The mean prediction error for the concrete deck PHM deterioration model using stationary transition probabilities is -0.36 with a standard deviation of 0.85 . The mean error of the concrete deck PHM deterioration model when using nonstationary transition probabilities is reduced to 0.02 with a standard deviation of 0.82 .

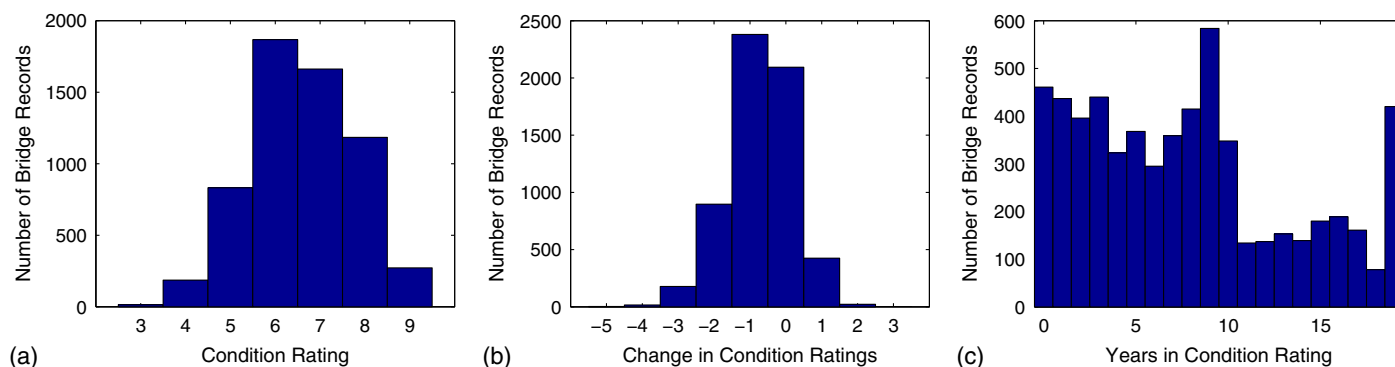


Fig. 8. Histograms of concrete deck condition rating data in observation period: (a) observed initial condition ratings; (b) change in condition ratings; and (c) pre-existing duration at initial observation.

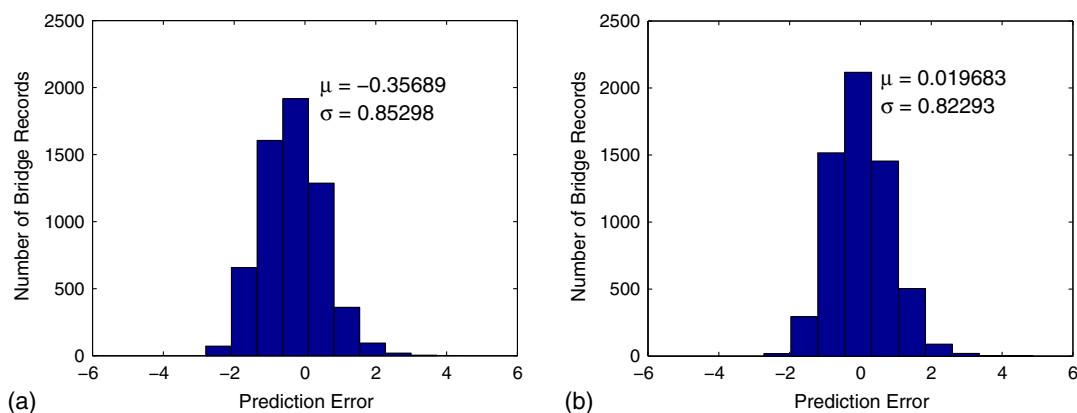


Fig. 9. Histograms of prediction errors of PHM bridge deterioration models applied to concrete deck condition rating data for a 15-year forecasting period: (a) stationary prediction errors; and (b) non-stationary prediction errors.

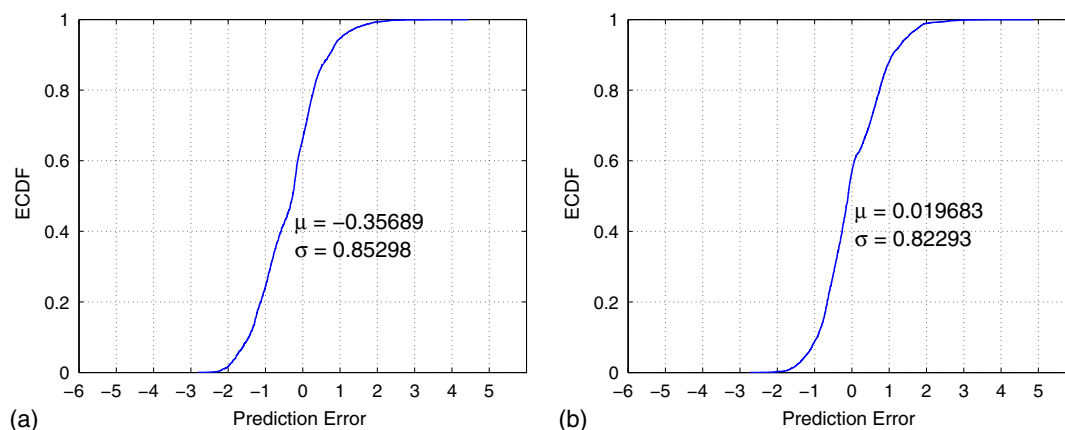


Fig. 10. Empirical cumulative distribution functions (ECDFs) of prediction errors of PHM bridge deterioration models applied to concrete deck condition rating data for a 15-year forecasting period: (a) stationary prediction errors; and (b) non-stationary prediction errors.

The authors could not find comparable data relating to quantitative assessment of predictive fidelity of duration-based models developed in prior studies. However, a comparative assessment of the conventional Markov chain models and linear-regression-based polynomial deterioration models is available in Jiang (2010). In that study, the assessment was made using a sample of 40 bridges from the Indiana bridge database over prediction periods of 2, 4, and 6 years from 1997 to 1999, to 2001, and to 2003, respectively. The study reported an average prediction error of 0.26 for the conventional Markov chain model with a standard deviation of 0.39, and an average error of 1.03 for the polynomial model with a standard deviation of 0.84 for the 6-year prediction period from 1997 to 2003. On comparing with the assessment results for the PHM deterioration models in the current study, it can be observed that the mean error from the stationary PHM deterioration models is comparable to the mean error from the conventional Markov chain model, while the mean error from the nonstationary PHM deterioration models is markedly lower. The larger standard deviation of errors from the PHM deterioration models may be attributed to the significantly larger sample size as well as the longer prediction period used in the current study relative to Jiang's Indiana study.

As an additional investigation to demonstrate the predictive capabilities of the PHM deterioration models presented in this paper, a further model assessment was performed on concrete deck records extracted from the North Carolina state database for the same prediction periods as reported in the Indiana study. The stationary PHM deterioration models exhibited mean prediction errors of -0.11 , -0.18 , and -0.24 over the prediction periods of 2, 4, and 6 years, respectively, which are noticeably lower than the mean prediction errors of 0.33, 0.45, and 0.26 reported over similar prediction periods in Jiang's Indiana study for the conventional Markov chain models.

The empirical cumulative distribution functions of the prediction errors from the stationary and nonstationary PHM deterioration models relative to the observed concrete deck condition ratings were also developed in the current study, and are presented in Fig. 10. These cumulative distribution functions are particularly useful for estimating the probability that the prediction will be within a prescribed bounded interval over the analyzed planning horizon. Because condition ratings are assigned on an integer scale, the interval of interest might correspond to ± 1 condition rating. It can be observed that this probability is about 70% for the stationary

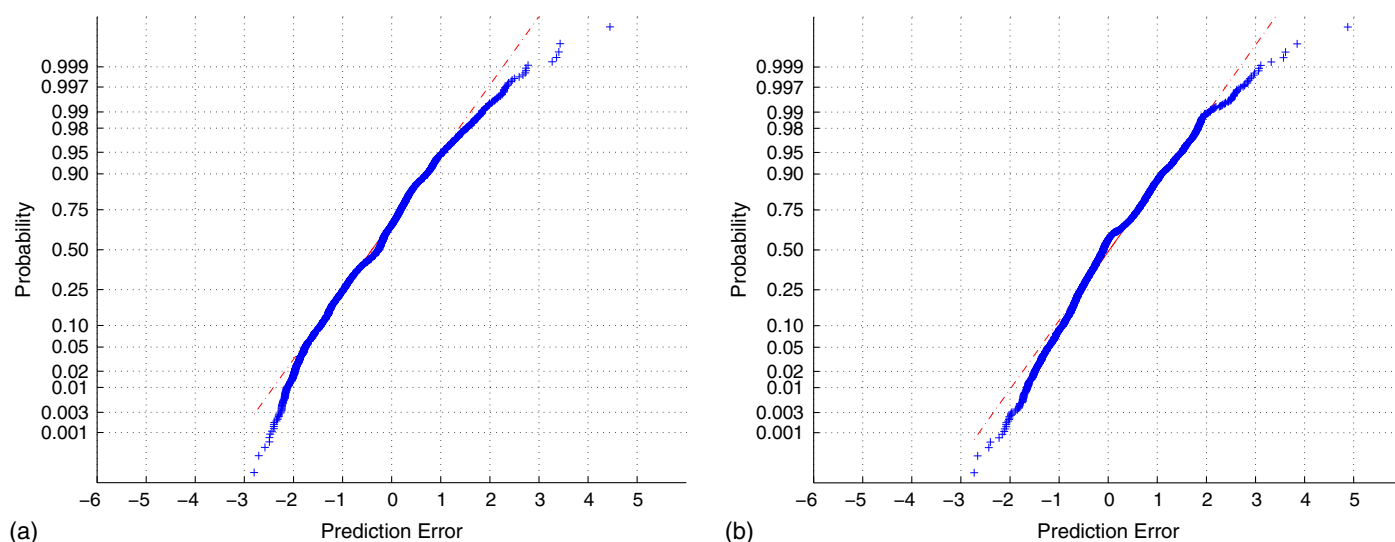


Fig. 11. Normal probability plots of prediction errors of PHM deterioration models applied to concrete deck condition rating data for a 15-year forecasting period: (a) stationary prediction errors; and (b) non-stationary prediction errors.

concrete deck PHM deterioration models. The probability that the nonstationary PHM concrete deck PHM deterioration model will estimate the actual condition rating within ± 1 over a 15-year planning horizon is close to 80%. Further, Fig. 11 presents the normal probability plots for the PHM deterioration model residuals. It can be observed that the error distributions for both stationary and nonstationary model types are fairly normally distributed, although the distribution for the nonstationary model appears to exhibit slightly less skew. Overall, the nonstationary PHM deterioration model exhibits the significant advantage of improved accuracy with an underlying robust design.

Conclusions

A complete duration-based multivariable probabilistic model for predicting bridge deterioration has been developed and successfully implemented on North Carolina's large statewide bridge database. The model uses a combination of the Cox proportional hazards survival analysis and the nonhomogeneous Markov chain theory. Proportional hazards analysis is used to statistically identify the most significant factors affecting the deterioration of a bridge component and quantify the effect of each factor at each condition rating along the life cycle of the component in terms of hazard ratios. An innovative approach to developing bridge-specific nonstationary transition probability matrices using PHM survival functions and hazard ratios was devised and the application in a discrete time nonhomogeneous Markov chain formulation for bridge deterioration was presented. The model is completely data driven, multivariable, duration dependent, and probabilistic, yet simple and straightforward in implementation. Its design is such that it can be integrated within the existing bridge management systems that employ Markov chain models for deterioration forecasting with minimal extension of the computational routines. An assessment of the predictive fidelity of the model using historical data demonstrated that with its capability to account for variation in deterioration rates over consecutive inspection cycles including the varying effects of influencing factors, the model achieves remarkable improvement in accuracy and precision compared to the conventional Markov chain models even over long-term planning horizons.

The model was implemented on NBI general condition rating data in this study in preference to element-level data owing to the 35-year-long duration of records available in the NBI database compared to only 10 years of element-level data. However, the advanced structure of the model is well suited for implementation on element-level data of increased granularity when sufficient duration of such data becomes available. Until then, the model can be effectively applied to take advantage of the long and rich database of bridge records available nationwide within the NBI database. The multivariable formulation of the model further allows for advantageously recording and investigating the significance of additional variables associated with bridge deterioration and provides a way of evaluating the comparative impact of maintenance interventions on the deterioration rate.

Acknowledgments

Most of the research presented in this paper was conducted as part of the research project RP 2014-07 sponsored by the North Carolina Department of Transportation. The financial support provided by this grant is greatly appreciated. The contents of this paper reflect the views of the authors and not necessarily the views of the NCDOT. The authors are responsible for the accuracy of the data presented herein. Additionally, this paper does not constitute

a standard, specification, or regulation and does not necessarily reflect official policies of NCDOT. In particular, the authors appreciate the guidance and technical support provided by Cary Clemmons.

References

- Agrawal, A. K., A. Kawaguchi, and Z. Chen. 2009. *Bridge element deterioration rates*. Final Rep. C-01-51. New York: New York State Dept. of Transportation.
- Agrawal, A. K., A. Kawaguchi, and Z. Chen. 2010. "Deterioration rates of typical bridge elements in New York." *J. Bridge Eng.* 15 (4): 419–429. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0000123](https://doi.org/10.1061/(ASCE)BE.1943-5592.0000123).
- Barbu, V., M. Boussemart, and N. Limnios. 2004. "Discrete-time semi-Markov model for reliability and survival analysis." *Commun. Stat. Theory Methods* 33 (11): 2833–2868. <https://doi.org/10.1081/STA-200037923>.
- Bulusu, S., and K. C. Sinha. 1997. "Comparison of methodologies to predict bridge deterioration." *Transp. Res. Rec.* 1597 (1): 34–42. <https://doi.org/10.3141/1597-05>.
- Butt, A., M. Y. Shahin, K. J. Feighan, and S. H. Carpenter. 1987. "Pavement performance prediction model using the Markov process." *Transp. Res. Rec.* 1123 (1): 12–19.
- Cinlar, E. 1975. *Introduction to stochastic processes*. Englewood Cliffs, NJ: Prentice Hall.
- Cox, D. R. 1972. "Regression models and life-tables." *J. R. Stat. Soc.* 34 (2): 187–202. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>.
- Cox, D. R., and D. Oakes. 1984. *Analysis of survival data*. London: Chapman and Hall.
- DeStefano, P. D., and D. A. Grivas. 1998. "Method for estimating transition probability in bridge deterioration models." *J. Infrastruct. Syst.* 4 (2): 56–62. [https://doi.org/10.1061/\(ASCE\)1076-0342\(1998\)4:2\(56\)](https://doi.org/10.1061/(ASCE)1076-0342(1998)4:2(56)).
- Duncan, S. A., and D. W. Johnston. 2002. *Bridge management system update*. Final Rep. FHWA/NC/2005-06. Raleigh, NC: Dept. of Civil Engineering, North Carolina State Univ.
- Fang, J., T. Ishida, and T. Yamazaki. 2018. "Quantitative evaluation of risk factors affecting the deterioration of RC deck slab components in East Japan and Tokyo regions using survival analysis." *Appl. Sci.* 8 (9): 1470. <https://doi.org/10.3390/app8091470>.
- Fleming, T. R., and D. P. Harrington. 1978. "Estimation for discrete time nonhomogeneous Markov chains." *Stochastic Processes Appl.* 7 (2): 131–139. [https://doi.org/10.1016/0304-4149\(78\)90012-1](https://doi.org/10.1016/0304-4149(78)90012-1).
- Frangopol, D. M., M.-J. Kallen, and J. M. Noortwijk. 2004. "Probabilistic models for life-cycle performance of deteriorating structures: Review and future directions." *Prog. Struct. Mater. Eng.* 6 (4): 197–212. <https://doi.org/10.1002/pse.180>.
- Golabi, K., and R. Shepard. 1997. "Pontis: A system for maintenance optimization and improvement of US bridge networks." *Interfaces* 27 (1): 71–88. <https://doi.org/10.1287/inte.27.1.71>.
- Goyal, R. 2015. "Development of a survival based framework for bridge deterioration modeling with large-scale application to the North Carolina bridge management system." Ph.D. dissertation, Dept. of Civil and Environmental Engineering, Univ. of North Carolina at Charlotte.
- Goyal, R., M. Whelan, and T. Cavalline. 2016. "Characterizing the effect of external factors on deterioration rates of bridge components using multivariable proportional hazards regression." *Struct. Infrastruct. Eng.* 13 (7): 894–905. <https://doi.org/10.1080/15732479.2016.1217888>.
- Greene, W. H. 1997. *Econometric analysis*. 3rd ed. Upper Saddle River, NJ: Prentice Hall.
- Hajnal, J. 1955. "The ergodic properties of non-homogeneous finite Markov chains." *Math. Proc. Cambridge Philos. Soc.* 52 (1): 67–77. <https://doi.org/10.1017/S0305004100030991>.
- Hatami, A., and G. Morcous. 2011. *Developing deterioration models for Nebraska bridges*. Final Rep. SPR-P1(11)M302. Lincoln, NE: Univ. of Nebraska–Lincoln.
- Hawk, H., and E. P. Small. 1998. "The BRIDGIT bridge management system." *Struct. Eng. Int.* 8 (4): 309–314. <https://doi.org/10.2749/101686698780488712>.

- Hearn, G. 2012. *Deterioration and cost information for bridge management*. Final Rep. CDOT-2012-4. Denver: Colorado Dept. of Transportation.
- Hosmer, D. W. J., and S. Lemeshow. 1999. *Applied survival analysis: Regression modeling of time to event data*. New York: Wiley.
- Huang, C.-C. 1977. "Non-homogeneous Markov chains and their applications." Ph.D. dissertation, Dept. of Mathematics, Iowa State Univ.
- Hyman, W. A., and D. J. Hughes. 1983. "Computer model for life-cycle cost analysis of statewide bridge repair and replacement needs." *Transp. Res. Rec.* 899 (1): 52–61.
- Jiang, Y. 2010. "Application and comparison of regression and Markov chain methods in bridge condition prediction and system benefit optimization." *J. Transp. Res. Forum* 49 (2): 91–110.
- Jiang, Y., M. Saito, and K. C. Sinha. 1988. "Bridge performance prediction model using the Markov chain." *Transp. Res. Rec.* 1180 (1): 25–32.
- Kalbfleisch, J. D., and R. L. Prentice. 1980. *The statistical analysis of failure time data*. Waterloo, ON, Canada: Wiley.
- Kallen, M. J., and J. M. van Noortwijk. 2005. "A study towards the application of Markovian deterioration processes for bridge maintenance modeling in the Netherlands." In *Proc., European Safety and Reliability Conf.: Advances in Safety and Reliability*, edited by K. Kolowrocki, 1021–1028. Rotterdam, Netherlands: A.A. Balkema.
- Kumar, D. 1995. "Proportional hazards modeling of repairable systems." *Qual. Reliab. Eng. Int.* 11 (5): 361–369. <https://doi.org/10.1002/qre.4680110507>.
- Kumar, D., and B. Klefsjö. 1994. "Proportional hazards model: A review." *Reliab. Eng. Syst. Saf.* 44 (2): 177–188. [https://doi.org/10.1016/0951-8320\(94\)90010-8](https://doi.org/10.1016/0951-8320(94)90010-8).
- Lawless, J. F. 1982. *Statistical models and methods for lifetime data*. Waterloo, ON, Canada: Wiley.
- Madanat, S., and W. H. W. Ibrahim. 1995. "Poisson regression models of infrastructure transition probabilities." *J. Transp. Eng.* 121 (3): 267–272. [https://doi.org/10.1061/\(ASCE\)0733-947X\(1995\)121:3\(267\)](https://doi.org/10.1061/(ASCE)0733-947X(1995)121:3(267)).
- Madanat, S., R. Mishalani, and W. H. W. Ibrahim. 1995. "Estimation of infrastructure transition probabilities from condition rating data." *J. Infrastruct. Syst.* 1 (2): 120–125. [https://doi.org/10.1061/\(ASCE\)1076-0342\(1995\)1:2\(120\)](https://doi.org/10.1061/(ASCE)1076-0342(1995)1:2(120)).
- Madanat, S. M., M. G. Karlaftis, and P. S. McCarthy. 1997. "Probabilistic infrastructure deterioration models with panel data." *J. Infrastruct. Syst.* 3 (1): 4–9. [https://doi.org/10.1061/\(ASCE\)1076-0342\(1997\)3:1\(4\)](https://doi.org/10.1061/(ASCE)1076-0342(1997)3:1(4)).
- Markow, M. J., and W. A. Hyman. 2009. *Bridge management systems for transportation agency decision making*. NCHRP Synthesis 397. Washington, DC: Transportation Research Board.
- Mauch, M., and S. Madanat. 2001. "Semiparametric hazard rate models of reinforced concrete bridge deck deterioration." *J. Infrastruct. Syst.* 7 (2): 49–57. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2001\)7:2\(49\)](https://doi.org/10.1061/(ASCE)1076-0342(2001)7:2(49)).
- Mishalani, R. G., and S. M. Madanat. 2002. "Computation of infrastructure transition probabilities using stochastic duration models." *J. Infrastruct. Syst.* 8 (4): 139–148. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2002\)8:4\(139\)](https://doi.org/10.1061/(ASCE)1076-0342(2002)8:4(139)).
- Moomen, M., Y. Qiao, B. R. Agbelie, S. Labi, and K. C. Sinha. 2016. *Bridge deterioration models to support Indiana's bridge management system*. Final Rep. FHWA/IN/JTRP-2016/03. West Lafayette, IN: Purdue Univ.
- Morcous, G., Z. Lounis, and M. S. Mirza. 2003. "Identification of environmental categories for Markovian deterioration models of bridge decks." *J. Bridge Eng.* 8 (6): 353–361. [https://doi.org/10.1061/\(ASCE\)1084-0702\(2003\)8:6\(353\)](https://doi.org/10.1061/(ASCE)1084-0702(2003)8:6(353)).
- Morcous, G., H. Rivard, and A. M. Hanna. 2002. "Modeling bridge deterioration using case-based reasoning." *J. Infrastruct. Syst.* 8 (3): 86–95. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2002\)8:3\(86\)](https://doi.org/10.1061/(ASCE)1076-0342(2002)8:3(86)).
- Nakat, Z. S., and S. M. Madanat. 2008. "Stochastic duration modeling of pavement overlay crack initiation." *J. Infrastruct. Syst.* 14 (3): 185–192. [https://doi.org/10.1061/\(ASCE\)1076-0342\(2008\)14:3\(185\)](https://doi.org/10.1061/(ASCE)1076-0342(2008)14:3(185)).
- Ng, S.-K., and F. Moses. 1998. "Bridge deterioration modeling using semi-Markov theory." In *Structural safety and reliability*, edited by N. Shiraishi, M. Shinozuka, and Y. K. Wen, 113–120. Rotterdam, Netherlands: A.A. Balkema.
- Osaki, S. 1985. *Stochastic system reliability modeling*. Philadelphia: World Scientific.
- Ravirala, V., and D. A. Grivas. 1995. "State increment method of life-cycle cost analysis for highway management." *J. Infrastruct. Syst.* 1 (3): 151–159. [https://doi.org/10.1061/\(ASCE\)1076-0342\(1995\)1:3\(151\)](https://doi.org/10.1061/(ASCE)1076-0342(1995)1:3(151)).
- Ross, S. M. 1970. *Applied probability models with optimization applications*. San Francisco: Holden-Day.
- Samrout, M., E. Chatelet, R. Kouta, and N. Chebbo. 2007. "Optimization of maintenance policy using the proportional hazard model." *Reliab. Eng. Syst. Saf.* 94 (1): 44–52. <https://doi.org/10.1016/j.res.2007.12.006>.
- Scherer, W. T., and D. M. Glagola. 1994. "Markovian models for bridge maintenance management." *J. Transp. Eng.* 120 (1): 37–51. [https://doi.org/10.1061/\(ASCE\)0733-947X\(1994\)120:1\(37\)](https://doi.org/10.1061/(ASCE)0733-947X(1994)120:1(37)).
- Sobanjo, J. O. 2011. "State transition probabilities in bridge deterioration based on Weibull sojourn times." *Struct. Infrastruct. Eng.* 7 (10): 747–764. <https://doi.org/10.1080/15732470902917028>.
- Sobanjo, J. O., and P. D. Thompson. 2001. *Development of agency maintenance, repair & rehabilitation (MR&R) cost data for Florida's bridge management system*. Final Rep. Contract No. BB-879. Tallahassee, FL: Florida Dept. of Transportation.
- Sobanjo, J. O., and P. D. Thompson. 2011. *Enhancement of FDOT's project level and network level bridge management analysis tools*. Final Rep. Contract No. BDK83 977-01. Tallahassee, FL: Florida Dept. of Transportation.
- Vassiliou, P.-C. G. 1998. "The evolution of the theory of non-homogeneous Markov systems." *Appl. Stochastic Models Data Anal.* 13 (3–4): 159–176. [https://doi.org/10.1002/\(SICI\)1099-0747\(199709/12\)13:3/4%3C159::AID-ASM309%3E3.0.CO;2-Q](https://doi.org/10.1002/(SICI)1099-0747(199709/12)13:3/4%3C159::AID-ASM309%3E3.0.CO;2-Q).
- Vassiliou, P.-C. G., and A. A. Papadopoulos. 1992. "Non-homogeneous semi-Markov systems and maintainability of state sizes." *J. Appl. Probab.* 29 (3): 519–534. <https://doi.org/10.2307/3214890>.
- Wang, K. C. P., J. Zaniewski, and G. Way. 1994. "Probabilistic behavior of pavements." *J. Transp. Eng.* 120 (3): 358–375. [https://doi.org/10.1061/\(ASCE\)0733-947X\(1994\)120:3\(358\)](https://doi.org/10.1061/(ASCE)0733-947X(1994)120:3(358)).
- Wellalage, N. K. W., T. Zhang, and R. Dwight. 2015. "Calibrating Markov chain-based deterioration models for predicting future conditions of railway bridge elements." *J. Bridge Eng.* 20 (2): 04014060. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0000640](https://doi.org/10.1061/(ASCE)BE.1943-5592.0000640).
- Yang, Y. N., M. M. Kumaraswamy, H. J. Pam, and H. M. Xie. 2013. "Integrating semiparametric and parametric models in survival analysis of bridge element deterioration." *J. Infrastruct. Syst.* 19 (2): 176–185. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000115](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000115).